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COMPUTER SECURITY WK2

1.SCENARIO ANALYSIS:

. Threat Scenario: DDoS Attack

A Distributed Denial-of-Service (DDoS) attack is a cybersecurity threat where many compromised computers or devices send a large number of requests to a targeted server or network. The purpose is to overwhelm the system's resources so that legitimate users cannot access its services. For example, an online shopping company could be targeted during a busy sales period. Attackers may use thousands of infected devices to send traffic to the company's website. As the traffic increases, the server becomes overloaded, causing the website to slow down or become completely unavailable. This can result in financial losses, customer dissatisfaction, and damage to the company's reputation. To reduce the risk of a DDoS attack, organizations can use firewalls, traffic filtering, rate limiting, load balancing, and DDoS protection services. Regular monitoring can also help detect unusual traffic patterns early and allow security teams to respond quickly.

2.CONCEPT RESEARCH:

Threat Research: Ransomware

Ransomware is a type of malware that encrypts a victim's files and prevents access until a ransom is demanded. It can seriously affect individuals, businesses, schools, and government organizations by causing data loss, financial damage, service interruptions, and loss of customer trust. Some ransomware attacks also steal sensitive information before encrypting files. To reduce the risk, organizations should regularly back up important data and keep backups offline or protected from unauthorized access. Systems and software should be updated to fix security vulnerabilities. Strong passwords, multi-factor authentication, antivirus software, firewalls, and employee cybersecurity training can also help prevent attacks. If an infection occurs, affected devices should be isolated immediately and professional cybersecurity assistance should be obtained.

3.TOOL PRACTICE:

Tool Practice: Wireshark Packet Analysis

Using Wireshark to analyze a network packet helped me understand how data travels across a network. I captured network traffic and examined packet information such as

source and destination IP addresses, protocols, packet length, and transmission details. I observed that different protocols are used for different communication tasks, such as TCP for reliable data transfer and DNS for converting domain names into IP addresses. This practical activity improved my understanding of network communication and cybersecurity monitoring. It also showed me how Wireshark can help identify unusual or suspicious traffic. Overall, the exercise strengthened my practical networking skills and demonstrated the importance of monitoring network activity for security purposes.

4.DIAGRAM

Explanation

This diagram shows how a DoS/DDoS attack is mitigated before it reaches the target server. Malicious traffic from multiple sources enters the internet and is first detected by the protection layer. The protection layer uses traffic filtering, rate limiting, and anomaly detection to block or absorb suspicious traffic. Only clean, legitimate traffic is allowed to pass through to the server, keeping services available. Continuous monitoring and alerts help detect attacks early and allow quick response. This approach reduces downtime, protects resources, and ensures users can access the system securely.



